Week 8

COGS 108 SP 2026

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Due dates

- A3: Thursday (05/21)
- D7: Friday (05/22)

Learning Modes & Classification Metrics

Before choosing a metric, identify the task and the error you care about.

Supervised Learning

- Use labeled data
- Learns relationship between features and known targets
- Goal is prediction/classification
- Has ground truth labels during training
- Examples: Linear Regression, Logistic Regression, SVM, Random Forest

Unsupervised Learning

- Use unlabeled data
- Finds hidden structure or patterns in data
- Goal is discovery, grouping, or dimensionality reduction
- No ground truth labels provided
- Examples: K-means, PCA, t-SNE, Hierarchical Clustering

Supervised learning learns from known answers.
Unsupervised learning tries to discover patterns without answers.

Linear vs. non-linear classifiers

Linear models

A machine learning method that separates different categories of data by drawing a straight line, flat plane, or flat hyperplane through the feature space.

E.g. Logistic Regression, Linear SVM, Perceptrons.
Stable, interpretable, and computationally inexpensive.

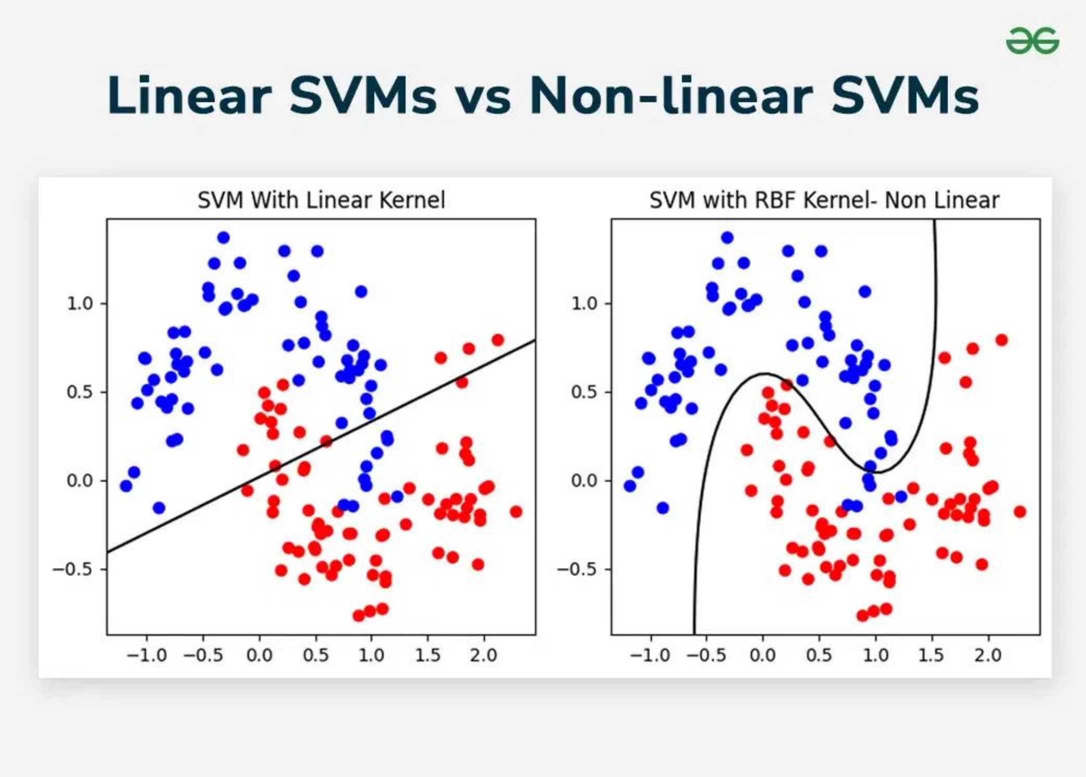
Non-Linear models

A machine learning method that separates data categories using curved lines or complex, irregular boundaries to capture intricate underlying patterns.

E.g. Kernel SVM, Random Forests, KNN, Deep Learning.
Powerful, flexible, and adaptive to data geometry.

Linear vs. non-linear classifiers

Different models can make very different boundaries on the same data.

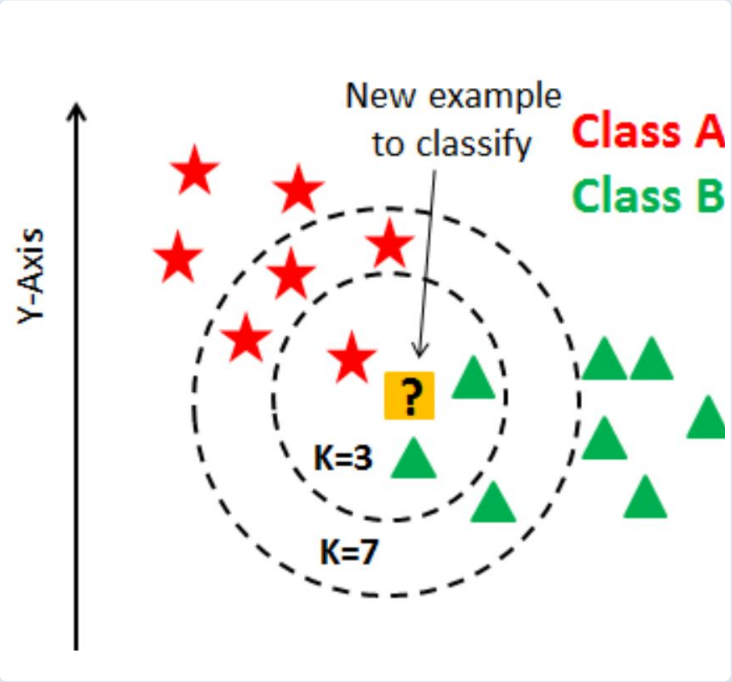


Selection principle

Use validation or cross-validation to estimate generalization.

kNN predicts by local voting

A new point is classified from the labels of its nearest training points.



Core idea

Measure distance from the new point to training points. Then use the nearby labels to make a prediction.

Main hyperparameter

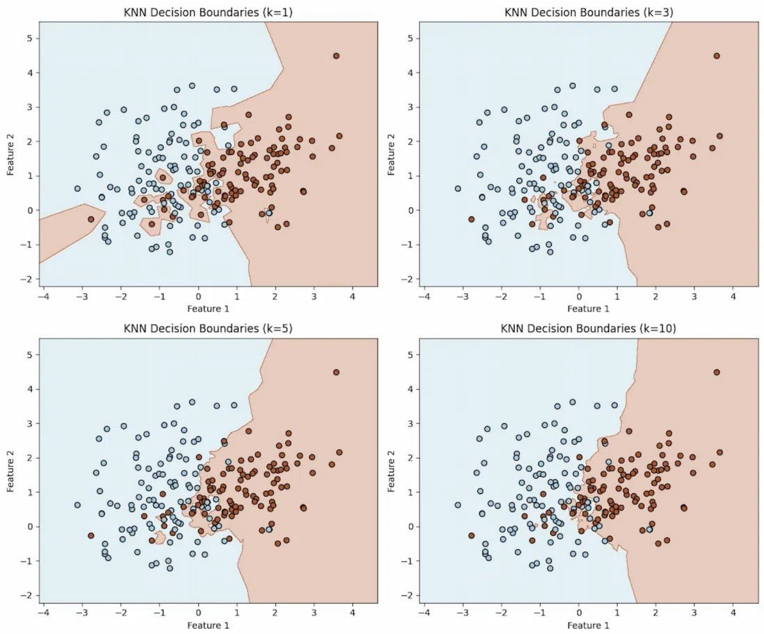
k controls how many neighbors vote. Small k is flexible. Large k is smoother.

Important preprocessing

Distance-based models care about scale, so numerical features often need standardization.

Changing k changes complexity

Small k can memorize local noise; larger k smooths the boundary and can underfit.



Lower k

More flexible, more variance, more noise-sensitive

Higher k

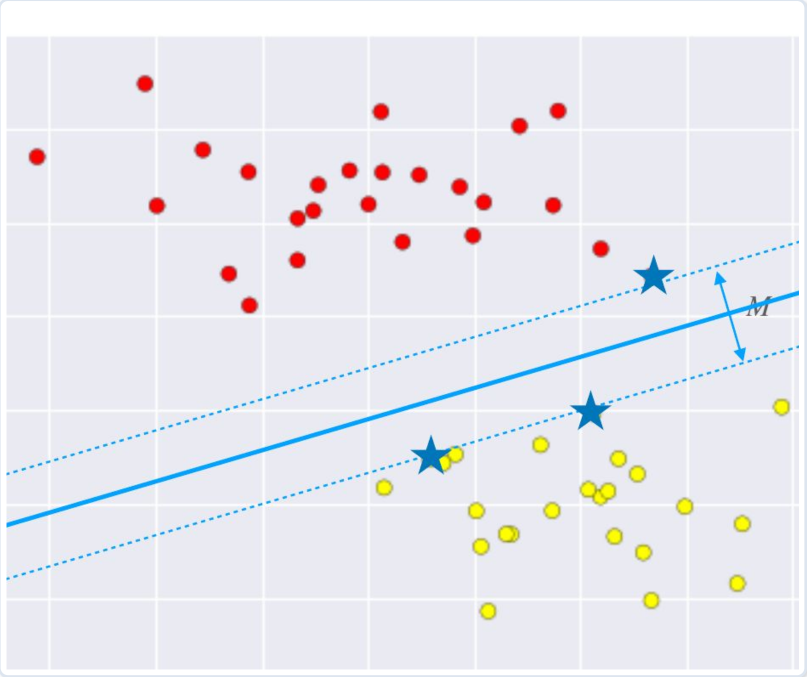
Smoother, Less variance, Risk of underfitting

More smoothing

Use validation performance, not just training performance, to choose k.

SVMs maximize the margin

The boundary is chosen to separate classes while keeping the nearest points as far away as possible.



Decision boundary

The separating line or plane. For a linear SVM, this boundary is linear in the feature space.

Margin

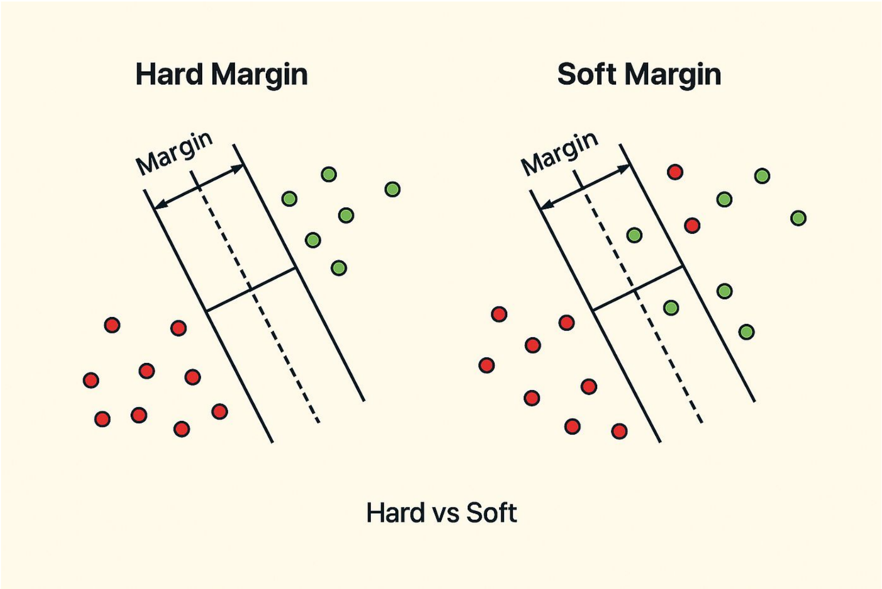
The empty space between the boundary and the closest training points.

Support vectors

The training points closest to the boundary. These points determine the margin.

C controls hard vs. soft margins

C changes how strongly the SVM tries to avoid training mistakes.



Large C

A harder margin. The model tries harder to classify training points correctly.

Small C

A softer margin. The model tolerates some mistakes to get a more robust boundary.

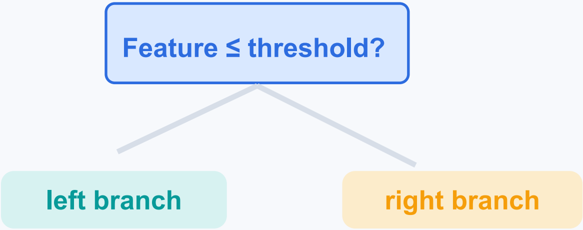
When data are noisy

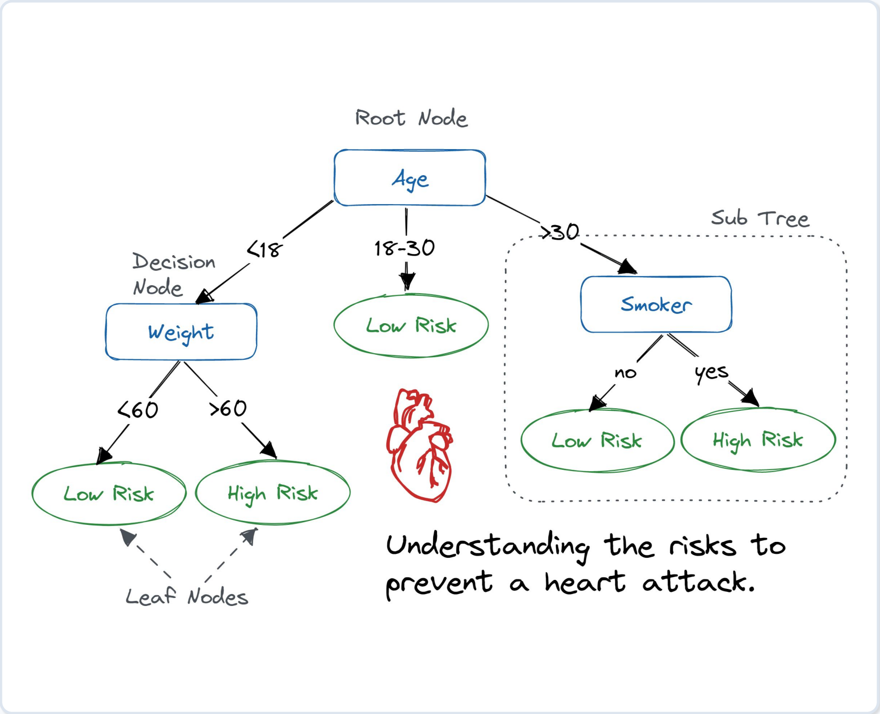
A softer margin often generalizes better than forcing a perfect boundary.

Decision trees are readable rules

A tree repeatedly splits data using if–then statements.

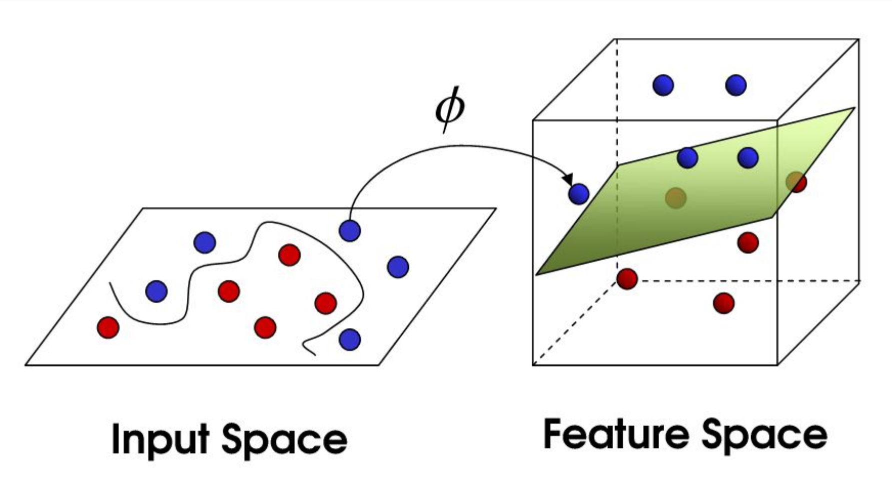
Each node asks a rule about one feature.
Each split creates axis-aligned decision boundaries.
The tree chooses splits that reduce impurity.
More depth means more flexible rules and more overfitting risk.





Kernels make linear models act non-linear

A kernel transforms the feature space so a linear separator can solve a non-linear problem.



Original space

The classes may not be separable by a straight line.

Higher-dimensional space

After transformation, a linear boundary can work.

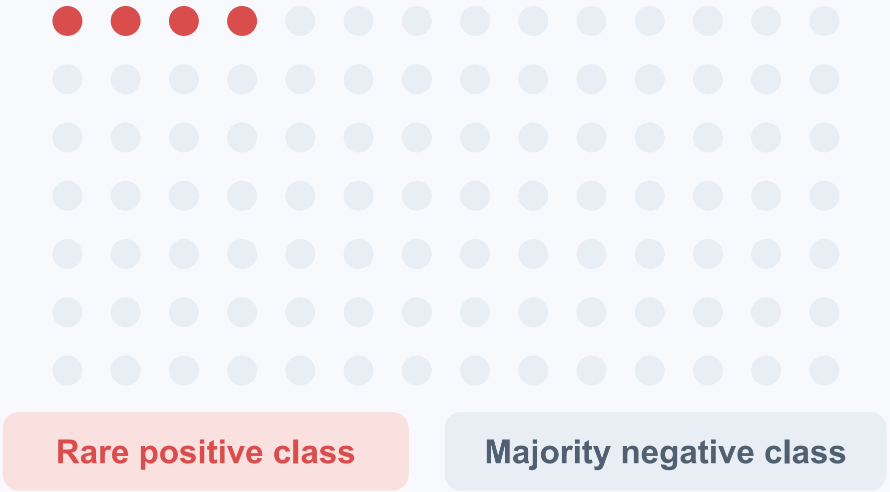
Kernel trick

The model can use this idea without explicitly building every transformed feature.

IMBALANCED DATA

Accuracy can mislead

When one class is rare, a high overall accuracy can hide failure on the class we care about.



Accuracy
What fraction of all predictions were correct?

Problem
A model can look accurate by always predicting the majority class.

Fix
Check class-specific metrics such as recall, specificity, and precision.

Start by defining the positive class

TP, FP, FN, and TN are defined relative to the class labeled “positive.”

		Predicted	
		+	–
Actual	+	TP True positive	FN False negative
	–	FP False positive	TN True negative

Four common metrics answer different questions

Accuracy

Of all examples, what percent were predicted correctly?

Specificity

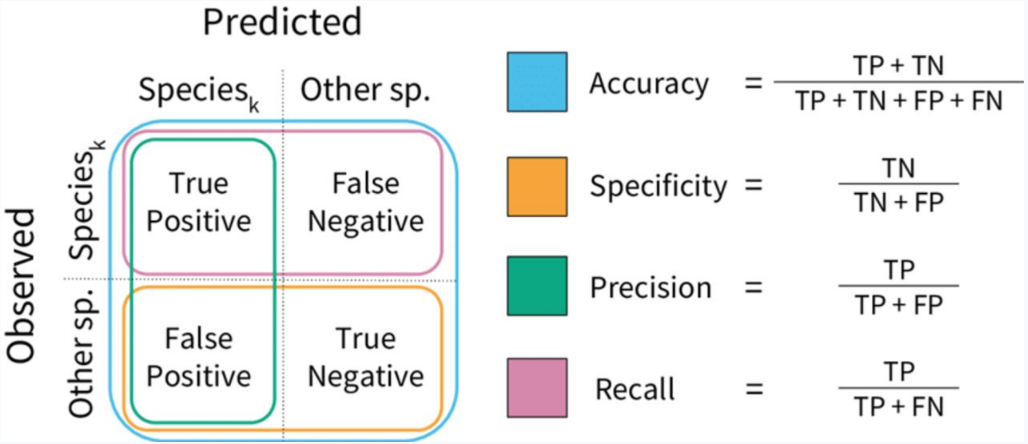
Of the actual negatives, what percent did we correctly reject?

Recall / Sensitivity

Of the actual positives, what percent did we detect?

Precision / PPV

Of the predicted positives, what percent were truly positive?



Shortcut: ask “What group is in the denominator?”

Continuous targets need error metrics

For regression, we compare predicted values against true values.

Mean squared error	$\text{MSE} = \frac{1}{n}\sum_{t=1}^{n} e_t^2$
Root mean squared error	$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} e_t^2}$
Mean absolute error	$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n} \lvert e_t \rvert$
Mean absolute percentage error	$\text{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n} \left\lvert \frac{e_t}{y_t} \right\rvert$

MSE averages squared errors

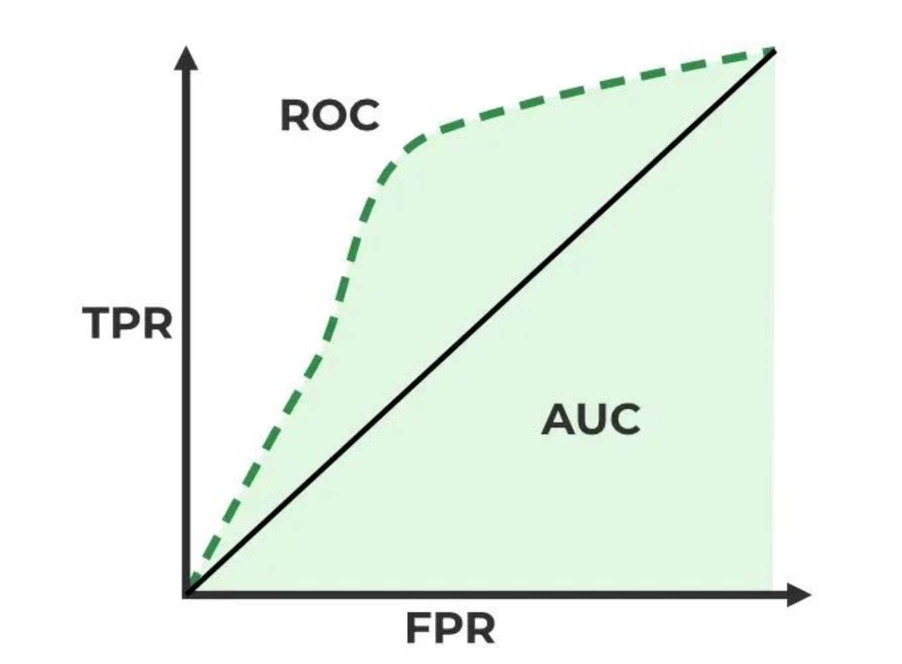
RMSE = square root of MSE

MAE averages absolute errors

Squared-error metrics punish large errors especially strongly. MAE is less dominated by extreme outliers.

ROC curves show threshold tradeoffs

AUC summarizes how well scores separate positives from negatives across thresholds.



ROC curve

Plots true positive rate against false positive rate as the threshold changes.

AUC

Area under the ROC curve. Higher values mean better separation.

Interpretation

Random-like performance is near 0.5 for balanced binary data. Perfect separation is 1.

Model selection prevents overfitting

Trying many models is allowed only if we evaluate them on data not used for fitting.

Algorithms

Try kNN, SVM, decision trees, neural nets, and others.

Hyperparameters

Try k, C, kernel type, max depth, and more.

Features

Try different predictor sets or engineered features.

Compare on validation data

If we tune using the test set, the test set stops being an honest estimate of future performance.

WORKFLOW

Separate fitting, choosing, and final testing

The validation set is for choosing a model; the test set is for the final performance estimate.

1. Shuffle and split

Create training, validation, and test data.

2. Fit candidates

Train each candidate on training data.

3. Choose and refit

Pick based on validation, then refit using training plus validation.

4. Final estimate

Evaluate once on the test data.

A reliable sklearn recipe

The notebook shows the practical ingredients: preprocessing, fitting, assessment, and selection.

Raw data
Features + target

Preprocess
standardize, one-hot

Fit model
kNN, SVM, tree

Evaluate
metrics + plots

Select
CV / grid search

Standardize continuous features when distance or magnitude matters.
One-hot encode unordered categorical variables.
Use train_test_split, cross_val_score / cross_val_predict, GridSearchCV, and Pipeline as needed.
Use grouped or stratified split strategies when observations are not independent or classes are imbalanced.

Main implementation principle: avoid leakage between training, validation, and test data.

Text features: tokens, TF–IDF, and sentiment

For text data, the model sees engineered representations of words or tokens.

“The penalty kick went over the bar.”

penalty | kick | went | over | bar

Tokenization

Split text into units such as words, terms, or n-grams.

Term frequency

How common a term is inside one document.

Inverse document frequency

Downweights terms that appear in many documents.

Sentiment analysis

Infer emotional tone or polarity from text.

Preprocessing choices matter: tokenizer, stop words, stemming or lemmatization, and how to handle rare/common terms.

Quiz Time

NO Phone, NO Communication, No Notes

Access Code: idk

Your time to …

- Work on projects
- Finish Checkpoint
- Finish D7 & A3
- Have fun…